

Module 03 — Causal Inference Analysis

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{r setup, include=FALSE} knitr::opts_chunk$set(echo = TRUE)
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Module 03 — Causal Inference Analysis

Part of the CausoBiome Pipeline

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Version: 1.0

Objective

This module identifies potentially causal microbial genes (ARGs/VFs) contributing to colorectal cancer (CRC) progression. It integrates feature annotation, functional distribution analysis, effect estimation using Double Machine Learning (DML), bootstrap robustness, interaction effects, and sensitivity analysis via E-values.

Prerequisite Notes

Prepare the annotation files for CARD and VFDB by extracting the following columns:

- **CARD:** Sample_ID, Gene, Matched_Species, AMR Gene Family, Resistance_Mechanism, Antibiotics_Class
- **VFDB:** Sample_ID, Gene, Matched_Species, Protein, Resistance_Mechanism, Class_Mechanism

These files are required before executing the causal analysis pipeline.

Scripts Included

01_Complex_heatmap_arg_vf.R

- **Purpose:** Generate z-score heatmap of consistent ARG/VF biomarkers with functional and taxonomic annotations.
 - **Inputs:**
 - combined_ARG_VFDB_final_DATA.csv
 - Metadata_Aligned_VF_ARG_CountMatrix.csv
 - Filtered_biomarker_annotated.csv
 - **Output:** heatmap_consistent_genes.png
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02_causal_analysis.py

- **Purpose:** End-to-end causal analysis and visualization of microbial gene effects on CRC stage.
- **Includes:**
 - Annotation and merging of robust biomarkers
 - Functional pie chart & group differences
 - Post-hoc Pearson residuals for functional shift
 - Gene-wise log2FC + statistical testing (Mann-Whitney + FDR)
 - Signature score analysis (mean & PCA-based)
 - Causal effect estimation (DML with Random Forest)
 - Bootstrap robustness (confidence intervals)
 - E-value sensitivity analysis
 - Interaction network effects and classification (synergistic, antagonistic)
- **Outputs:**
 - robust_biomarkers_annotated.csv, Filtered_biomarker_annotated.csv
 - heatmap_log2fc_consistent_biomarkers.png
 - DML_Causal_Effects_Per_Species.csv, Sensitivity_Analysis_Evalues.csv
 - Significant_Microbial_Interactions_Final.png, Gene_Interaction_Network_Metrics.csv

Output Summary

Analysis Step	Key Outputs
Functional Annotation	Filtered_biomarker_annotated.csv
Expression Heatmap	heatmap_consistent_genes.png
Functional Shift Analysis	Functional_Group_Differences_Across_Stages.csv
log2FC + Stats	statistical_comparison_log2fc.csv, heatmap_log2fc_...
Signature Score Profiling	signature_scores.csv, boxplots
Causal Estimation (ATE)	DML_Causal_Effects_Per_Species.csv
Bootstrap Robustness	Bootstrap_Causal_Effects_CI.csv, Bootstrap_Robustness_Check.png
E-value Sensitivity	Sensitivity_Analysis_Evalues.csv, evalute plot
Interaction Effects	Microbial_Interaction_Effects_Corrected.csv, Significant_Microbial_Interactions_Final.png
Interaction Network & Metrics	Gene_Interaction_Network_Metrics.csv

End of Module 03 README